MAKERERE UNIVERSITY

College of Computing and Information Sciences (CoCIS)

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SMART CONTROL AND MONITORING SYSTEM

FOR THE COLLEGE OF COMPUTING AND INFORMATION SCIENCES

Project Requirements and Design Specification

ESP32, ESP-IDF and Wokwi Embedded Systems Coursework

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1. Introduction

1.1 Purpose of this document

This document is the single reference specification for the Smart Control and Monitoring System for the College of Computing and Information Sciences (CoCIS), Makerere University. It consolidates every requirement stated in the coursework brief, proposes innovative additions, defines the system and software architecture, lists the hardware and the embedded systems concepts to be demonstrated, and sets out the plan, test strategy and submission checklist. It is written so that team members and AI coding agents can use it as a shared and unambiguous source of truth while the system is designed, implemented, simulated and documented.

1.2 Scope

The document covers the pilot deployment described in the brief: one main entrance with separate entry and exit sections, and two lecture rooms. It covers firmware for three ESP32 terminals developed with ESP-IDF and simulated on Wokwi, together with the data, communication and security design that connects them. It also covers the academic deliverables (report, simulation link, individual videos, contribution statements and presentation).

1.3 Intended audience and how to use it

Table 1: Audience and recommended reading

Reader	Use of this document	Start with
Team members	Understand the whole system, choose tasks, and prepare for questions during the presentation.	Sections 2, 6, 7, 19, 20
Hardware and simulation lead	Build the Wokwi circuits, pin maps and part substitutions.	Sections 8, 17
Firmware developers	Implement tasks, state machines, drivers and protocols.	Sections 9 to 13, 15, 16
Report and video authors	Map every implemented feature to a requirement and an embedded systems concept.	Sections 4, 10, 18, 20
AI coding agents	Generate and review code against stable requirement IDs, pin maps, topics and schemas.	Section 22 first, then 4, 8, 11, 12

1.4 Conventions

- **Priority keywords** follow the MoSCoW scheme: **Must** (mandatory, drawn from the brief or needed for it to work), **Should** (strongly recommended, forms the innovation score), **Could** (optional if time allows).

- **Requirement identifiers** have the form FR-xxx-nn (functional), NFR-nn (non-functional) and INN-nn (innovative feature). Tests use TC-nn. Identifiers never change once issued, so code comments, commits and the report can cite them.
- **Source of truth:** where this document and the coursework brief disagree, the brief wins and this document must be corrected. Items marked as assumptions should be confirmed with the lecturer.
- **Terminology:** the terms node, terminal and ESP32 are used interchangeably. The Entrance node is also called the Hub because it owns the authoritative data.

2. Project Overview

2.1 Background and problem statement

CoCIS houses valuable equipment that needs reliable protection against unauthorised access. The college also incurs high electricity costs, partly because lights are left on in unoccupied rooms. Management agreed in early October 2026 to build an automated system, first as a pilot covering two lecture rooms and the main entrance. The system must identify people with RFID tags, record entries and exits, know who is inside at any time, switch lights according to occupancy, enforce a configurable access period, store data on a memory card, and expose information through a serial interface, all coordinated across several ESP32 microcontrollers without data races.

2.2 Objectives

1. Prevent unauthorised building access through RFID identification and a configurable access schedule.
2. Maintain an accurate, auditable record of every entry and exit and a live view of who is inside.
3. Reduce electricity waste by switching lecture room lights automatically according to occupancy.
4. Provide reliable coordination between three ESP32 terminals with proper synchronisation.
5. Give management a secure serial interface for monitoring, configuration and decision-making.
6. Demonstrate a deep understanding of embedded systems design, supported by innovative features.

2.3 Scope of the pilot

Table 2: In scope and out of scope

In scope	Out of scope (documented as future work)
Three ESP32 terminals: Lecture Room 1, Lecture Room 2, Main Entrance	Roll-out to all college rooms and floors
Separate ENTRY and EXIT RFID readers and door mechanisms	Real RFID card enrolment hardware and printing

In scope	Out of scope (documented as future work)
PIR-based room occupancy and relay-based lamp control	Mains-voltage wiring, electrical certification
SD card logging, serial CLI, authenticated configuration	Mobile app and cloud analytics back end
Wokwi simulation using ESP-IDF	Physical prototype (simulation only is required)

2.4 Constraints

- Microcontroller: ESP32 only. Framework: ESP-IDF with FreeRTOS. Simulator: Wokwi.
- Team size: at most five members. Every member must be able to explain the code they claim.
- Submission deadline: **2 November 2026**, on Muele. Only students who attend the presentation receive marks.
- Available time from the date of this document (27 October 2026) is six days, so scope must be managed with the priorities in Section 6.

2.5 Assumptions

- Wokwi simulation access to the internet through the Wokwi-GUEST Wi-Fi network is available for MQTT and SNTP. This must be verified on day one (Section 17).
- Real RFID readers are replaced in simulation by whatever the platform supports (Section 8.4). The firmware isolates the reader behind a driver interface so the choice does not affect the rest of the design.
- A fixed registry of 10 to 20 sample users is enough for the demonstration.
- The time zone is East Africa Time (UTC+3) with no daylight saving.

2.6 Deliverables and assessment alignment

Table 3: Deliverables from the course work guide.

#	Deliverable	Content	Assessment link
1	System Operation Report	Architecture, hardware, software design, implemented features	Quality and completeness of report
2	Wokwi simulation link	Public or shared link to the working project(s)	Working demonstration
3	Individual video demonstrations	Each member explains the system and the code they wrote	Clarity of explanation of own contribution

#	Deliverable	Content	Assessment link
4	Individual contributions	Tasks and responsibilities of each member	Accountability
5	Group member details	Names and identification details	Identification
6	Presentation to lecturer	Live demo and questions to every member	Marks only for attendees; deductions for unexplained claims

> **Marking emphasis:** the course work guide states that particular emphasis is placed on innovation and additional features. The innovation plan in Section 6 is therefore a core part of the project and not an optional extra.

3. Stakeholders and System Actors

Table 4: Actors and their privileges

Actor	Description	Privileges in the system
Registered user (staff, student)	Holds an RFID tag, enters and leaves through the main entrance.	Scan at ENTRY and EXIT readers within the access period.
Visitor	Holds a temporary tag with an expiry time (innovation).	Same as user but only until the card expires.
Security officer	Monitors the building and responds to alerts.	View status, occupancy, logs, roll-call; trigger emergency mode; suspend a tag.
Administrator	Manages the configuration and users.	All security officer rights plus schedule changes, user management, settings, password changes.
College management	Consumes reports for decisions.	Read-only reports: access summaries, energy saving estimates.
Maintenance technician	Services hardware.	Diagnostics view (node health, sensor values).
Lecturer (assessor)	Evaluates the project.	Observes the demonstration and asks questions.

4. Functional Requirements

Each table lists the requirements for one area of the brief. The final column cites the section of the coursework brief that the requirement comes from, or marks it as derived (needed to make the brief work) or as an addition.

4.1 Access control and identification

Table 5: Access control requirements

ID	Requirement	Priority	Source
FR-ACC-01	The main entrance has two designated sections, ENTRY and EXIT. Each has its own RFID reader, door actuator, status LEDs and feedback.	Must	Brief 2.1
FR-ACC-02	Every person must scan an RFID tag at the ENTRY reader to enter and at the EXIT reader to leave.	Must	Brief 2.1
FR-ACC-03	On a valid tag the system verifies identity against the user registry and, if all rules pass, activates the corresponding door mechanism for a configurable time (default 5 s).	Must	Brief 2.1
FR-ACC-04	On a rejected tag the door stays locked, the red LED and buzzer signal denial, and the reason code is shown and logged.	Must	Derived
FR-ACC-05	Every scan outcome is recorded with timestamp, UID, user name, direction, result and reason.	Must	Brief 2.1
FR-ACC-06	Duplicate reads of the same tag within 2 s are ignored (debounce).	Should	Derived
FR-ACC-07	Anti-passback: entry is refused if the tag is already recorded inside. An exit with no recorded entry is granted (in hours) but flagged as an anomaly.	Should	Addition
FR-ACC-08	The registry stores per user: UID, name, role, status (ACTIVE, SUSPENDED, EXPIRED) and optional expiry date.	Must	Derived
FR-ACC-09	Three consecutive denied scans from the same unknown or suspended UID raise a security alert.	Should	Addition

4.2 Occupancy monitoring

Table 6: Occupancy requirements

ID	Requirement	Priority	Source
FR-OCC-01	The Hub maintains the building occupancy count and the set of UIDs currently inside.	Must	Brief 2.1
FR-OCC-02	Management can list who is inside at any given time (name, UID, time of entry).	Must	Brief 2.1
FR-OCC-03	Occupancy state survives a reboot or power loss (persisted to NVS and SD).	Must	Derived

ID	Requirement	Priority	Source
FR-OCC-04	The count can never become negative; inconsistencies are corrected and logged.	Should	Derived
FR-OCC-05	Each lecture room reports an occupied or vacant status derived from its sensor.	Must	Brief 2.2, 2.5
FR-OCC-06	A roll-call report (who is still inside) is produced at closing time and on emergency.	Should	Addition

4.3 Automated lighting control

Table 7: Lighting requirements

ID	Requirement	Priority	Source
FR-LGT-01	Each lecture room node detects whether its room is occupied using a presence sensor.	Must	Brief 2.2
FR-LGT-02	Lights switch on when the room is occupied and off after a configurable vacancy timeout (default 300 s; 20 s in demo mode).	Must	Brief 2.2
FR-LGT-03	Lighting status and occupancy are reported to the Hub and visible on the CLI.	Must	Brief 2.5
FR-LGT-04	A pre-off warning (lamp blinks) is given before lights switch off, so a still person can wave and restart the timer.	Should	Addition
FR-LGT-05	Daylight gating: if an LDR reports enough ambient light the lamp is not switched on.	Should	Addition
FR-LGT-06	A manual override button forces lights on for a limited time (default 60 min).	Should	Addition
FR-LGT-07	Energy accounting: lamp-on time per room and estimated saving versus an always-on baseline.	Should	Addition
FR-LGT-08	Room nodes keep working autonomously if the Hub or network is unavailable.	Must	Derived (reliability)
FR-LGT-09	Motion in a room while the building is recorded empty or closed raises an intrusion alert and keeps lights off.	Should	Addition

4.4 Distributed control with ESP32 terminals

Table 8: Distributed control requirements

ID	Requirement	Priority	Source
FR-DST-01	The system consists of three terminals: Lecture Room 1, Lecture Room 2 and Main Entrance, each on its own ESP32.	Must	Brief 2.3
FR-DST-02	Terminals exchange events, status and configuration over a defined messaging protocol (Section 11).	Must	Brief 2.3
FR-DST-03	Each node sends a heartbeat every 5 s; a node silent for 15 s is flagged offline, and its Last Will message is honoured.	Should	Addition

ID	Requirement	Priority	Source
FR-DST-04	A node that reconnects resynchronises: it receives the retained configuration and uploads any buffered events.	Should	Addition
FR-DST-05	The same room firmware runs on Room 1 and Room 2, differing only by a configured ROOM_ID.	Should	Derived (maintainability)

4.5 Data storage and management

Table 9: Storage requirements

ID	Requirement	Priority	Source
FR-STO-01	An SD card module (SPI, FAT filesystem) stores access logs, occupancy snapshots, lighting and energy records, user registry and schedule.	Must	Brief 2.4
FR-STO-02	Logs are written as CSV, one file per day, with an append-only policy and periodic flush.	Must	Brief 2.4
FR-STO-03	Only one task (storage_task) writes to the SD card; all other tasks send records through a queue.	Must	Brief 2.7
FR-STO-04	If the SD card is missing or fails, records are kept in a RAM ring buffer, an alert is raised, and buffered records are flushed when the card returns.	Should	Derived
FR-STO-05	Each log record carries a SHA-256 hash chained to the previous record to make tampering detectable; a verify command checks the chain.	Should	Addition
FR-STO-06	Configuration (schedule, credentials, thresholds) is persisted in NVS flash and mirrored on SD.	Must	Derived

4.6 Serial interface and management menus

Table 10: Serial interface requirements

ID	Requirement	Priority	Source
FR-CLI-01	Users can query the system through text commands on the serial interface (UART0, 115200 baud).	Must	Brief 2.5
FR-CLI-02	The interface provides access records, current building occupancy, room occupancy, lighting status and system information.	Must	Brief 2.5
FR-CLI-03	A help menu lists commands available to the logged-in role; unknown commands produce a helpful message.	Must	Brief 2.5
FR-CLI-04	Configuration commands (schedule, users, thresholds) require login with an authorised role (Section 14).	Must	Brief 2.6
FR-CLI-05	Additional output interfaces are provided: OLED summary at the Hub, status LEDs, MQTT status topics.	Should	Brief 2.5 (additional)
FR-CLI-06	Sessions expire after 120 s of inactivity and password input is not echoed.	Should	Addition

4.7 Configurable building access schedule

Table 11: Schedule requirements

ID	Requirement	Priority	Source
FR-SCH-01	Management can configure the permitted access period (open and close time) at any time without reflashing firmware.	Must	Brief 2.6
FR-SCH-02	Outside the configured period both the ENTRY and EXIT doors stay locked for normal scans.	Must	Brief 2.6
FR-SCH-03	People still inside when the period ends are not allowed to leave through normal scans; they are listed and security is alerted (see assumption A-1 in Section 16).	Must	Brief 2.6
FR-SCH-04	Only authenticated users with the ADMIN role can modify the schedule; every change is logged with who, when, old value and new value.	Must	Brief 2.6
FR-SCH-05	The schedule supports a separate window per weekday and a list of holiday dates (closed).	Should	Addition
FR-SCH-06	A schedule change takes effect on all nodes consistently using versioned distribution (Section 13).	Must	Brief 2.7
FR-SCH-07	If valid time is not available the system fails secure: entries are denied and an alert is raised.	Should	Derived
FR-SCH-08	Emergency mode overrides the schedule and unlocks both doors (life safety).	Should	Addition

4.8 Resource synchronisation and reliability

Table 12: Synchronisation requirements

ID	Requirement	Priority	Source
FR-SYN-01	Shared data (occupancy table, schedule, SD card, SPI bus) is protected by FreeRTOS mutexes with timeouts and priority inheritance.	Must	Brief 2.7
FR-SYN-02	Cross-node updates follow a single-writer rule: only the Hub modifies authoritative records; rooms send requests or events.	Must	Brief 2.7
FR-SYN-03	Every event has a unique ID (node, boot count, sequence) so duplicate deliveries are detected and ignored (idempotency).	Must	Brief 2.7
FR-SYN-04	Configuration has a monotonically increasing version; updates with a stale base version are rejected.	Must	Brief 2.7
FR-SYN-05	A leased distributed lock (Section 13) serialises operations that need exclusive multi-node access.	Should	Brief 2.7
FR-SYN-06	Task watchdog and health supervision restart or flag a stuck task or node.	Should	Derived
FR-SYN-07	Mutex acquisition order is fixed (cfg, state, sd) to prevent deadlock.	Must	Derived

5. Non-Functional Requirements

Table 13: Non-functional requirements with measurable targets

ID	Category	Requirement and target
NFR-01	Performance	Scan to door-actuation latency at the Hub under 1 s; scan event visible on MQTT within 2 s.
NFR-02	Performance	PIR trigger to lamp on under 1 s; no missed motion events at 1 Hz sampling.
NFR-03	Reliability	No unhandled crash during a 30 minute soak test with scripted random scans; watchdog resets must recover to a consistent state.
NFR-04	Reliability	After a forced reboot the occupancy list, schedule and users are identical to before (FR-OCC-03, FR-STO-06).
NFR-05	Security	No plain-text passwords in code, NVS or logs; salted hash only. Lockout after 3 failed logins for 60 s.
NFR-06	Safety	Emergency egress is always possible (emergency mode unlocks doors; the exit door is fail-safe by design).
NFR-07	Maintainability	Modular components with one responsibility each, documented public headers, no magic numbers (Kconfig or constants).
NFR-08	Resource use	At least 30 percent of heap free at steady state; stack high-water mark checked for every task; no dynamic allocation inside ISRs.
NFR-09	Usability	Every CLI command has help text; LED and buzzer patterns follow one documented table (Section 8.5).
NFR-10	Portability	Hardware access only through a HAL so a real MFRC522 or a simulated reader can be swapped by configuration.
NFR-11	Observability	ESP_LOG levels per module; a diagnostics command prints heap, stack watermarks, queue depths and uptime.
NFR-12	Documentation	Every source file has a header comment naming its author, purpose and the requirement IDs it implements.

6. Innovative Features

The brief awards particular emphasis to innovation. The features below were chosen because each one improves security, efficiency, reliability or usefulness to management and each one also demonstrates a distinct embedded systems concept. They are ranked by value for effort so that the team can stop at any point and still hold a coherent system.

6.1 Innovation catalogue

Table 14: Proposed innovative features

ID	Feature	Benefit to management	Concept demonstrated	Tier	Effort
INN-01	Anti-passback and tailgating-safe occupancy	Prevents card sharing; keeps the inside list truthful	State machines, mutex-protected tables	1	S
INN-02	Emergency evacuation mode with live roll-call	Life safety; shows who is still inside	ISR, event groups, priority	1	S
INN-03	Hash-chained tamper-evident logs	Audit trail that cannot be silently altered	mbedtls SHA-256, FATFS, single-writer queue	1	M
INN-04	Intrusion detection by cross-checking RFID occupancy with PIR motion	Detects entry without a valid scan	Sensor fusion, distributed state	1	M
INN-05	Daylight-aware lighting (LDR gating)	Extra energy savings	ADC, thresholds with hysteresis	1	S
INN-06	Pre-off warning blink and manual override timer	Avoids switching lights off on still occupants	Software timers, FSM	1	S
INN-07	Energy analytics: kWh and UGX saved per room	Quantifies the business case	Counters, time accounting, NVS	2	M
INN-08	Offline-first operation and store-and-forward	Keeps working when Wi-Fi or broker fails	Ring buffers, reconnect logic, idempotency	1	M
INN-09	Role-based access, salted hashing, lockout, session timeout	Protects configuration	Crypto, RBAC, state handling	1	M
INN-10	Closing-time sweep: list and alert for people still inside	Supports security patrol	Esp_timer scheduling, alerts	2	S
INN-11	Visitor temporary cards with expiry	Flexible guest management	Date handling, registry	2	S
INN-12	Tailgating detection with a passage sensor	Detects two people on one scan	GPIO ISR counting	3	M

ID	Feature	Benefit to management	Concept demonstrated	Tier	Effort
INN-13	Node health monitoring (heartbeat, LWT, diagnostics)	Early fault detection	MQTT LWT, watchdogs	2	S
INN-14	Two-factor physical admin (admin card plus PIN keypad)	Stronger configuration security	Keypad scan, debounce	3	M
INN-15	Peak occupancy statistics per hour	Helps plan opening hours	Aggregation, memory budgeting	3	S

Tier 1 features are the recommended minimum innovation set for the demonstration. Tier 2 should be added once Tier 1 is stable. Tier 3 features are stretch goals and may be described in the report as future work. Effort: S is under half a day, M is about one day for one person.

6.2 Detailed description of Tier 1 features

INN-01 Anti-passback

The Hub stores a per-user inside flag. An ENTRY scan for a user already marked inside is refused with DENY_ANTIPASSBACK, which stops one tag being passed back to a second person. An EXIT scan without a recorded entry is allowed within hours but logged as ANOMALY_EXIT_NO_ENTRY and flagged for review, so nobody is trapped by a data error.

INN-02 Emergency evacuation mode

A dedicated emergency button (GPIO interrupt) or the CLI command sets the EMERGENCY bit in an event group. The Hub unlocks both doors, switches all lights on through a retained emergency message and records every pass as EMERGENCY_PASS. The roll-call report then compares the inside list against the people who passed the exit. Because the emergency path has the highest task priority it cannot be starved by logging or network activity.

INN-03 Hash-chained logs

Each log line stores hash = SHA256(previous_hash + line_content). Changing or deleting one line breaks every later hash, and the command log verify reports the first broken record. This brings a security and audit property typically seen in blockchains into a few dozen lines of embedded code and is easy to demonstrate by editing the CSV and running the verify command.

INN-04 Intrusion detection

Each room node publishes motion events. The Hub compares them with the occupancy count and schedule. If a room reports motion while the building is recorded empty or closed, or after the access period has ended without any recorded entry, an INTRUSION_SUSPECTED alert is raised, the room lights stay off, and the CLI and buzzer announce it. The detection relies on cooperation between three nodes, which makes it a good distributed-systems example.

INN-05 and INN-06 Intelligent lighting

The lighting state machine (Section 9.6) adds daylight gating and a visible warning blink before switching off. Both reduce wasted energy without annoying occupants. The warning window and vacancy timeout are configurable, which also allows a short demonstration timeout.

INN-08 Offline-first operation

The Hub decides access locally using its own registry and schedule, so a Wi-Fi or broker failure never blocks the door. Room nodes keep controlling lights from their local sensor. Events that could not be published are held in a bounded ring buffer and replayed in order after reconnection, with unique IDs so the Hub can discard duplicates.

INN-09 Authentication and authorisation

Credentials are stored as PBKDF2-HMAC-SHA256 hashes with a per-user salt. Roles (ADMIN, SECURITY, VIEWER) restrict commands. Three failed logins lock the account for 60 s, sessions expire after inactivity, and every login and configuration change is written to the hash-chained audit log.

7. System Architecture

7.1 Architecture overview

The system follows a hub-and-spoke distributed architecture. Three ESP32 terminals are connected through Wi-Fi to an MQTT broker. The Entrance node acts as the Hub: it owns the authoritative user registry, occupancy table, access schedule and SD card, and it coordinates configuration updates. The two Lecture Room nodes are autonomous controllers that handle sensors and lamps locally and report to the Hub. Figure 1 shows the complete system.

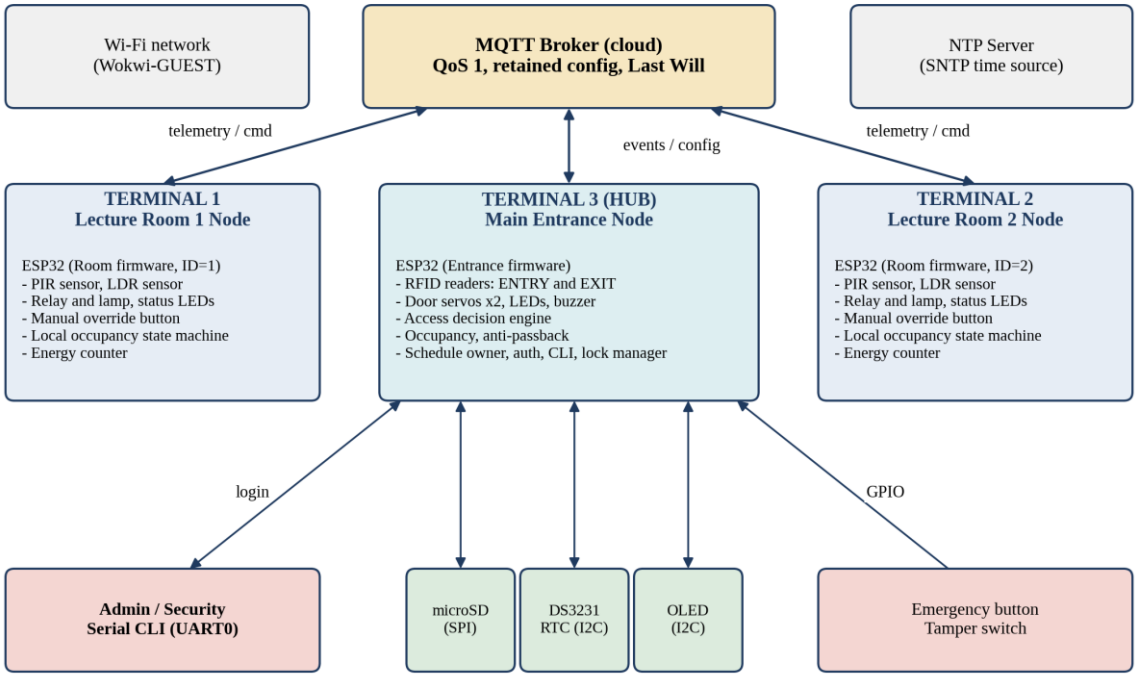


Figure 1: System architecture of the CoCIS smart control and monitoring system

7.2 Responsibilities of each terminal

Table 15: Node responsibilities

Node	Responsibilities	Owns (authoritative)
Terminal 3: Entrance (Hub)	ENTRY and EXIT RFID handling, access decision, door actuation, occupancy and anti-passback, SD logging, schedule enforcement, authentication, serial CLI, emergency handling, distributed lock and config coordination, OLED display.	User registry, occupancy table, access log, schedule and settings, audit log
Terminal 1: Lecture Room 1	PIR and LDR sampling, lighting state machine, relay control, local override, energy counters, telemetry, heartbeat, offline buffering.	Its own sensor state and lamp state (reported, not shared for writing)

Node	Responsibilities	Owns (authoritative)
Terminal 2: Lecture Room 2	Same firmware as Terminal 1 with ROOM_ID = 2.	Its own sensor state and lamp state

7.3 Key design decisions

Table 16: Design decisions and rationale

Decision	Rationale	Alternative considered
Central authority at the Hub (single writer)	Removes most race conditions by construction; simplifies consistency of occupancy and schedule.	Peer-to-peer replication (complex consensus, unnecessary for three nodes)
MQTT over Wi-Fi as the inter-node bus	Publish/subscribe fits event-driven design; supports QoS, retained messages and Last Will; works with Wokwi-GUEST.	ESP-NOW (not supported by the Wokwi simulator), raw TCP sockets (more code)
Room nodes decide lighting locally	Lights keep working during network failure; low latency.	Hub-driven lighting (single point of failure)
SD card only at the Hub	Central log and registry; cheaper; rooms keep short RAM ring buffers.	SD on every node (more hardware, more sync)
Hardware abstraction layer for every device	Lets the team swap the RFID source and re-use drivers; helps AI agents generate code.	Direct driver calls in application code
Fail-secure entry, fail-safe emergency	Protects equipment while guaranteeing a way out in an emergency.	Always-locked doors (unsafe)

7.4 Deployment topology on Wokwi

The brief requires one ESP32 per terminal, so the baseline deployment uses three Wokwi projects, one per terminal, communicating through a public MQTT broker reached over the Wokwi-GUEST Wi-Fi network. Each project holds the full circuit for its node. To avoid cross-talk with other student groups on a public broker, every topic starts with a unique group prefix (Section 11).

Table 17: Deployment options

Option	Description	When to use
A (baseline)	Three Wokwi projects, three ESP-IDF firmwares, MQTT between them. The lecturer opens the three links; run them side by side.	Default if the day-one spike (Section 17) shows MQTT works between two simulations.
B (fallback)	One Wokwi project containing the three terminals as separate task groups on one ESP32, linked by an in-memory message bus that mirrors the MQTT topic and payload design.	Only if multi-project MQTT or the broker is unreachable. Report must state the limitation honestly.

> **Verification required:** at the time of writing the exact Wokwi capabilities (multiple boards in one project, network access, available RFID and SD parts) must be confirmed hands-on. Section 17 defines the spike tests that settle this on day one. The architecture is intentionally transport-neutral so that the decision does not change the application code.

7.5 Layered software architecture

Each node uses the same four-layer structure shown in Figure 2. Application modules depend only on services; services depend on drivers (HAL); drivers depend on ESP-IDF. No application code touches registers or GPIOs directly.

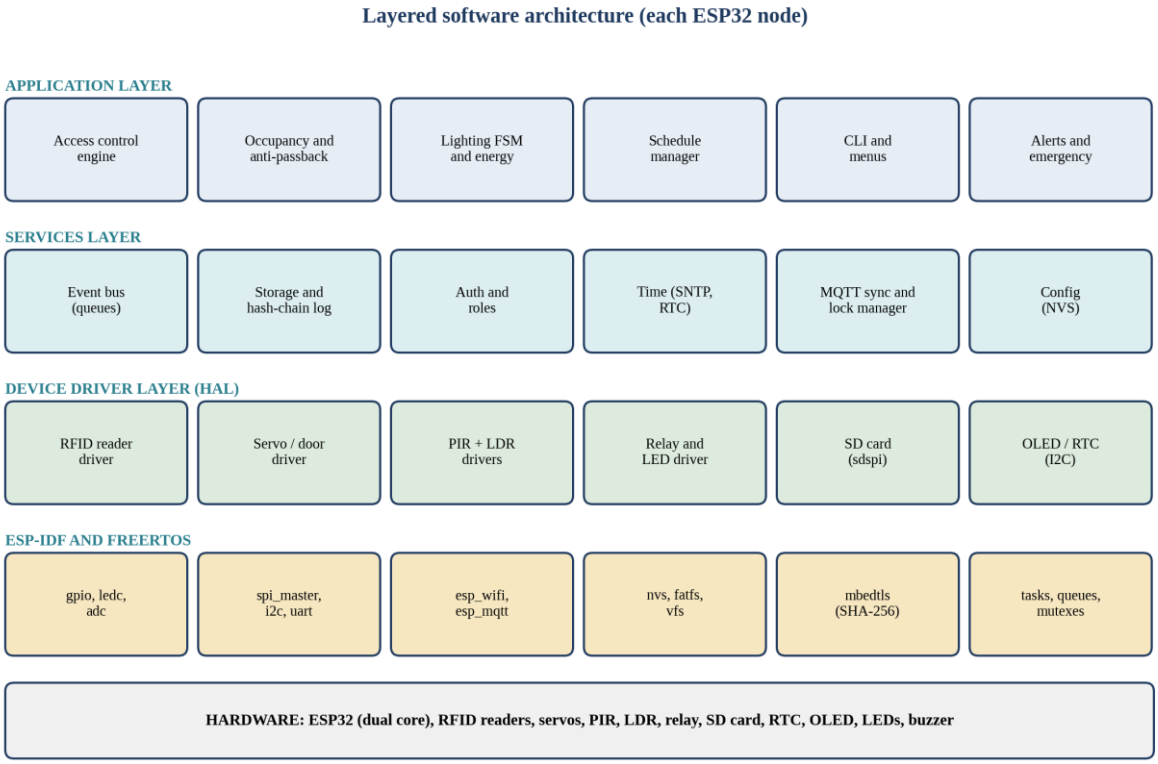


Figure 2: Layered software architecture used on every node

7.6 Operating modes

Table 18: System operating modes

Mode	Condition	Behaviour
NORMAL	Inside the access period, network up	Standard access control, occupancy, lighting, logging.
AFTER_HOURS	Outside the access period	ENTRY and EXIT scans denied; lights off unless override; motion triggers intrusion check; closing sweep list available.
EMERGENCY	Emergency button or CLI command	Both doors unlocked, lights on, roll-call, high-priority alert; schedule ignored.

Mode	Condition	Behaviour
DEGRADED	Network, broker or SD failure	Local decisions continue; events buffered; alert raised; retry with back-off.
MAINTENANCE	Admin sets maintenance flag	Doors locked normally, test commands enabled, access logged as TEST.

8. Hardware Design

8.1 Bill of components

Table 19: Components required (simulation parts in Wokwi)

Component	Qty	Node	Purpose	Wokwi part (verify)
ESP32 DevKit	3	All	Microcontroller, Wi-Fi	ESP32 DevKit V1
RFID reader (13.56 MHz)	2	Hub	ENTRY and EXIT identification	See Section 8.4
RFID tags/cards	10 to 20	Demo	User identities (UIDs)	Simulated UIDs
Servo motor SG90	2	Hub	Door latch mechanisms (0 deg locked, 90 deg open)	Servo
microSD card module	1	Hub	Logs, registry, configuration	microSD card
RTC DS3231	1	Hub	Battery-backed time fallback	DS3231 / DS1307 RTC
OLED SSD1306 128x64	1 (+2 optional)	Hub (rooms)	Local status display	SSD1306
PIR motion sensor HC-SR501	2	Rooms	Room occupancy detection	PIR motion sensor
LDR light sensor module	2	Rooms	Daylight measurement	Photoresistor sensor
Relay module	2	Rooms	Switch the lamp circuit	Relay module
Lamp (LED as stand-in)	2	Rooms	Room lights	LED
Status LEDs (red, green, yellow, blue)	4 (hub) + 6 (rooms)	All	Feedback and status	LED
Resistors 220 ohm	10+	All	LED current limiting	Resistor
Buzzer	1	Hub	Audible feedback and alarm	Buzzer
Push buttons	4	Hub (2), rooms (1 each)	Emergency, tamper, manual override	Pushbutton

8.2 Entrance Hub wiring and pin map

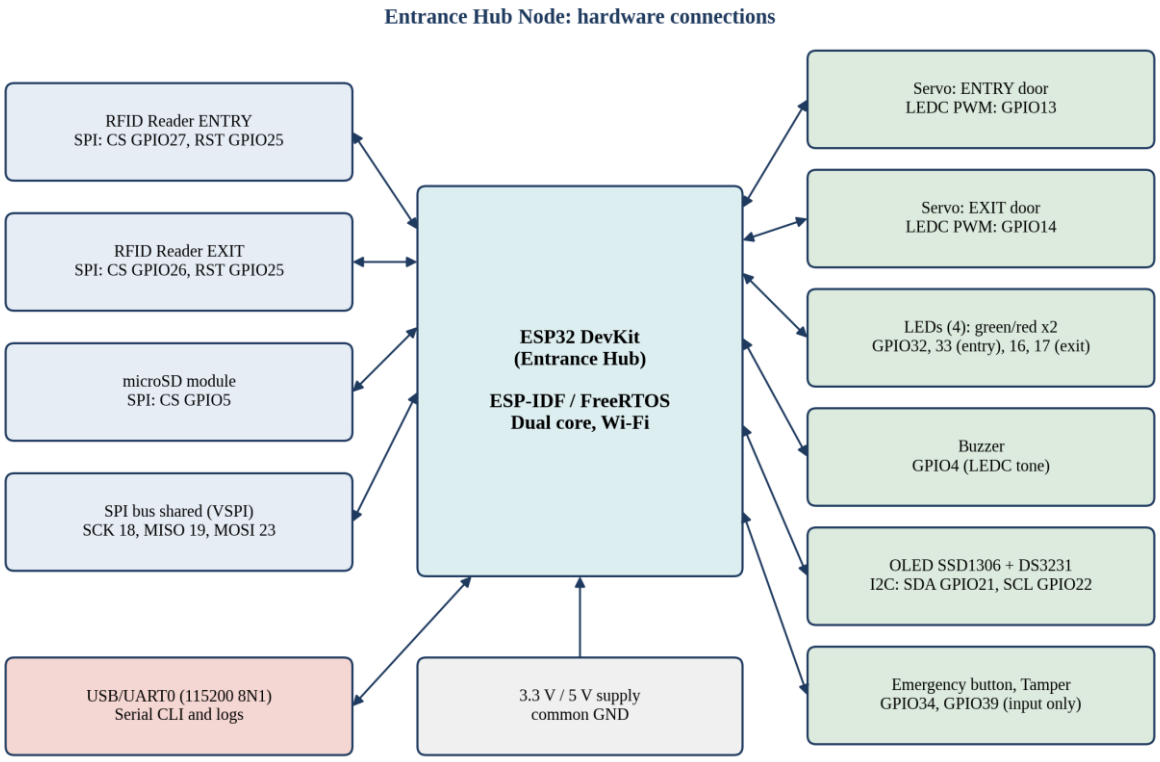


Figure 3: Entrance Hub hardware connections

Table 20: Hub GPIO assignment

Signal	GPIO	Type	Notes
SPI SCK / MISO / MOSI (VSPI)	18 / 19 / 23	SPI bus	Shared by SD card and both RFID readers; guarded by sd_mutex
SD card CS	5	Output	Idle high
RFID ENTRY CS	27	Output	Idle high
RFID EXIT CS	26	Output	Idle high
RFID RST (shared)	25	Output	
I2C SDA / SCL	21 / 22	I2C	OLED (0x3C) and DS3231 (0x68)
Servo ENTRY door	13	LEDC PWM 50 Hz	1 ms = locked, 2 ms = open (calibrate)
Servo EXIT door	14	LEDC PWM 50 Hz	
LED ENTRY green / red	32 / 33	Output	
LED EXIT green / red	16 / 17	Output	
Buzzer	4	LEDC tone	
Emergency button	34	Input (ISR, falling edge)	GPIO 34 to 39 are input only and have no internal pull-ups: use an external pull-up

Signal	GPIO	Type	Notes
Tamper switch	39	Input (ISR)	External pull-up
Passage sensor (optional, INN-12)	35	Input (ISR)	External pull-up
UART0 (CLI)	1 / 3	UART	115200 8N1 through USB

8.3 Lecture room wiring and pin map

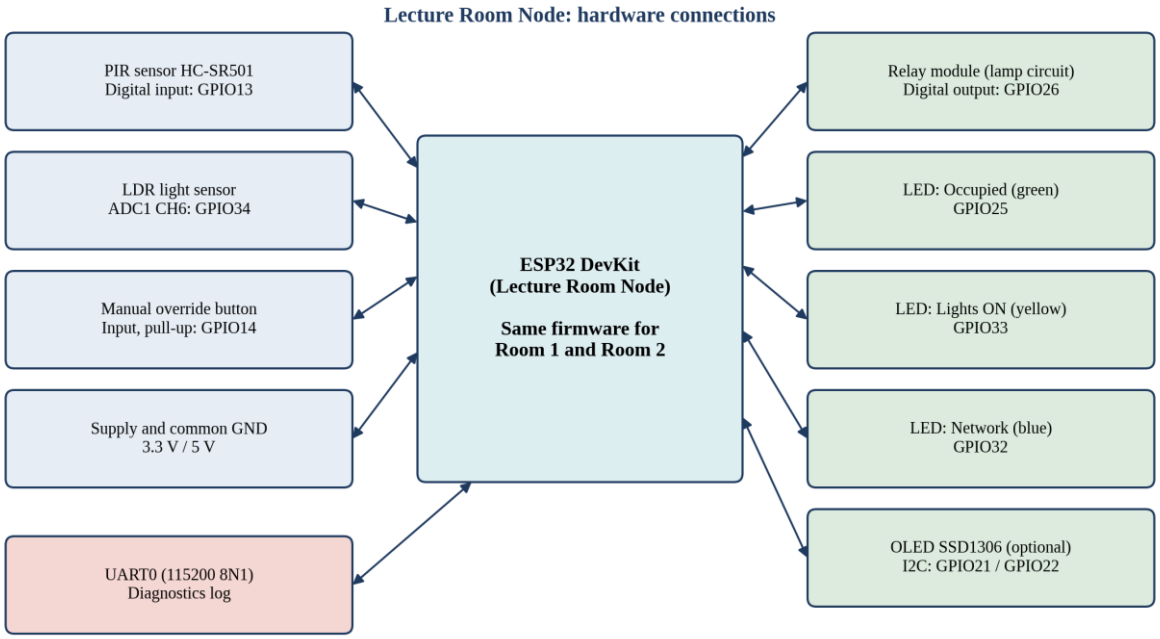


Figure 4: Lecture room node hardware connections

Table 21: Room node GPIO assignment (identical on Room 1 and Room 2)

Signal	GPIO	Type	Notes
PIR output	13	Input	HC-SR501 retrigger mode; 3.3 V logic
LDR analog out	34	ADC1 channel 6	Use ADC1 only (ADC2 conflicts with Wi-Fi)
Manual override button	14	Input, pull-up	Active low, debounced
Relay IN (lamp)	26	Output	Active high in Wokwi; document polarity
LED Occupied (green)	25	Output	
LED Lights ON (yellow)	33	Output	
LED Network (blue)	32	Output	Blinks while connecting, steady when linked
I2C SDA / SCL (optional OLED)	21 / 22	I2C	

> **GPIO cautions:** avoid strapping pins (0, 2, 12, 15) for outputs that affect boot. GPIO 5 is a strapping pin but is acceptable as an idle-high chip select. GPIO 6 to 11 are connected to flash and must not be used.

8.4 Wokwi part availability and substitutions

Table 22: Simulation risks for specific parts

Part	Risk	Planned substitute
RFID reader	The simulator may not offer a ready RFID reader part.	Keep the HAL interface rfid_read_uid(reader_id). Provide two implementations selected in Kconfig: (1) hardware driver for MFRC522 over SPI for realism; (2) simulated reader fed by console commands (sim scan entry UID) or a Wokwi custom chip. Decide on day one.
Door mechanism	No door part exists.	Servo for latch plus LED or label in the report; angle shown in the log.
Lamp	No mains lamp.	LED driven through the relay contact circuit.
PIR sensor	Motion is simulated by clicking the part.	Use hold-time logic to filter bounce; include a scripted demo sequence.
SD card	Needs a formatted image in the project.	Use the Wokwi SD card part with an empty FAT image; create folders at boot if missing.

8.5 Feedback semantics

Table 23: LED, buzzer and OLED feedback table

Event	Green LED	Red LED	Buzzer	OLED
Access granted	On 2 s	Off	1 short beep	Welcome, name, direction
Denied	Off	On 2 s	2 long beeps	Denied and reason code
Outside hours	Off	Blink 3 times	3 short beeps	Closed until HH:MM
Emergency	Both doors green	Off	Continuous slow pulse	EMERGENCY ROLL-CALL
Intrusion alert	Off	Fast blink	Fast beeps	INTRUSION room N
SD or network fault	Off	Slow blink	Single beep per minute	DEGRADED

9. Software Design

9.1 ESP-IDF project organisation

Each node is an ESP-IDF project. Shared code lives in components that are copied or symlinked into every project so that the same file implements the same behaviour on all nodes. The structure below is the agreed baseline.

Table 24: Repository and component layout

Path	Content
common/cocis_types	Shared enums, reason codes, event structs, message schema constants, error codes.
common/cocis_proto	JSON encode and decode, topic builders, event ID generation.
common/cocis_log_chain	SHA-256 chained log record builder and verifier.
common/cocis_time	SNTP, RTC fallback, schedule time helpers, EAT conversion.
entrance_node/main	app_main, task creation, board configuration.
entrance_node/components/hal_rfid, hal_door, hal_oled, hal_rtc, hal_sd	Device drivers for the Hub.
entrance_node/components/access_ctrl, occupancy, schedule, auth, cli, storage, net_sync, emergency	Application and service modules for the Hub.
room_node/main	app_main with ROOM_ID from Kconfig.
room_node/components/hal_pir, hal_ldr, hal_relay, hal_led	Device drivers for rooms.
room_node/components/light_fsm, energy, net_sync, diag	Application and service modules for rooms.
docs/	This document, report drafts, diagrams, test logs, Wokwi links.

9.2 Hub task architecture

The Hub uses preemptive FreeRTOS multitasking on both cores. Time-critical RFID and door tasks run on core 1; Wi-Fi, MQTT and CLI tasks are pinned to core 0 together with the ESP-IDF network stack. Tasks communicate by queues and event groups; shared state is touched only inside short mutex-protected sections (Figure 5).

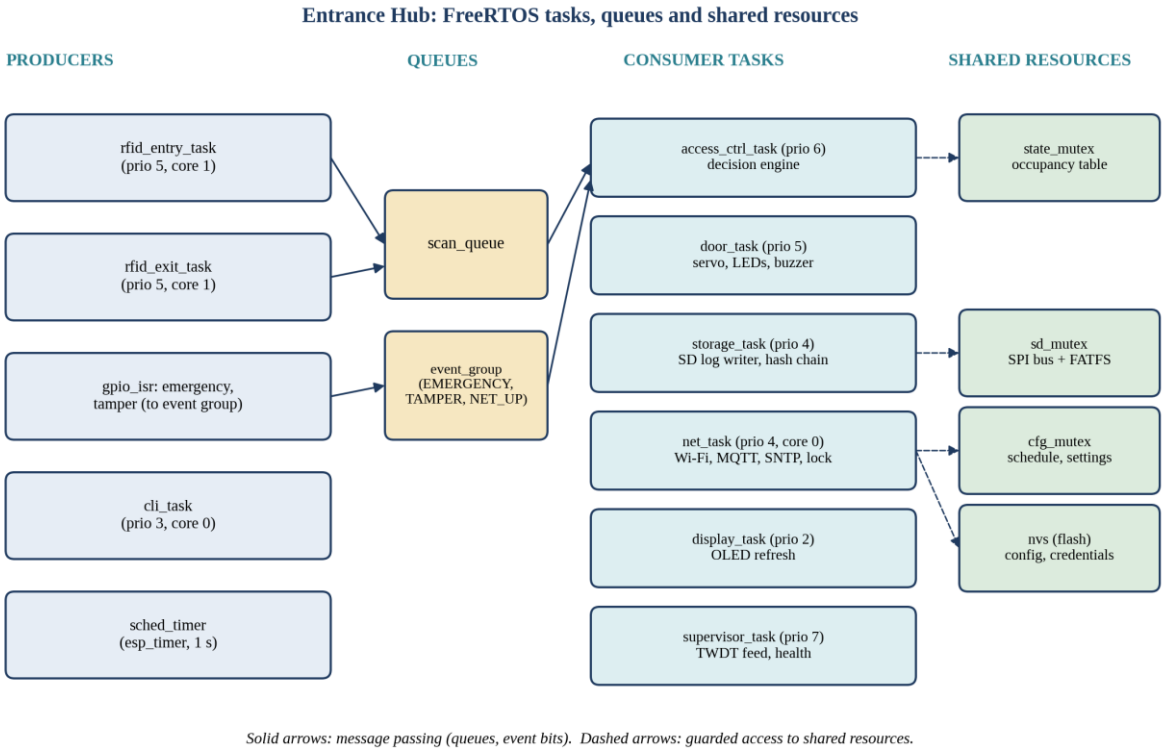


Figure 5: Entrance Hub tasks, queues and shared resources

Table 25: Hub task table

Task	Priority	Stack (bytes)	Responsibility
supervisor_task	7	3072	Feeds the task watchdog, monitors queue depths, heap, heartbeats; raises DEGRADED mode.
access_ctrl_task	6	4096	Consumes scan_queue, runs the decision algorithm, updates occupancy, posts door, log and network commands.
rfid_entry_task, rfid_exit_task	5	3072	Poll or interrupt-driven read of each reader, debounce, post scan events.
door_task	5	2560	Drives servos, LEDs and buzzer; enforces open time; fail-secure relock.
storage_task	4	6144	Sole SD writer; builds chained log records; flush policy; rotation; ring-buffer drain.
net_task	4	6144	Wi-Fi, MQTT client, SNTP, heartbeat, lock coordinator, config distribution.
cli_task	3	6144	Line editor, parser, session and role handling, command execution.
display_task	2	3072	Refreshes OLED every 500 ms from a snapshot of state.

Priorities follow the principle that safety and access decisions outrank logging and display. Stack sizes are initial values to be tuned with uxTaskGetStackHighWaterMark.

9.3 Room node task architecture

Table 26: Room task table

Task	Priority	Stack (bytes)	Responsibility
light_fsm_task	5	3072	Runs the lighting state machine every 100 ms from sensor events and timers; drives relay and LEDs.
sensor_task	4	2560	PIR edge handling with hold time; LDR sampling and averaging at 2 Hz.
net_task	4	6144	Wi-Fi, MQTT, config receive, telemetry, heartbeat, offline buffer.
energy_task	3	2560	Accumulates lamp-on seconds and estimated energy; persists every 5 min.
diag_task	2	2560	Prints diagnostics and feeds watchdog.

9.4 Access decision algorithm

Figure 6 specifies the order of checks executed by the access control task for every scan. The order matters: cheap and safety-critical checks come first, and all state changes happen in one short critical section after every rule has passed, so a half-applied decision can never be observed by another task.

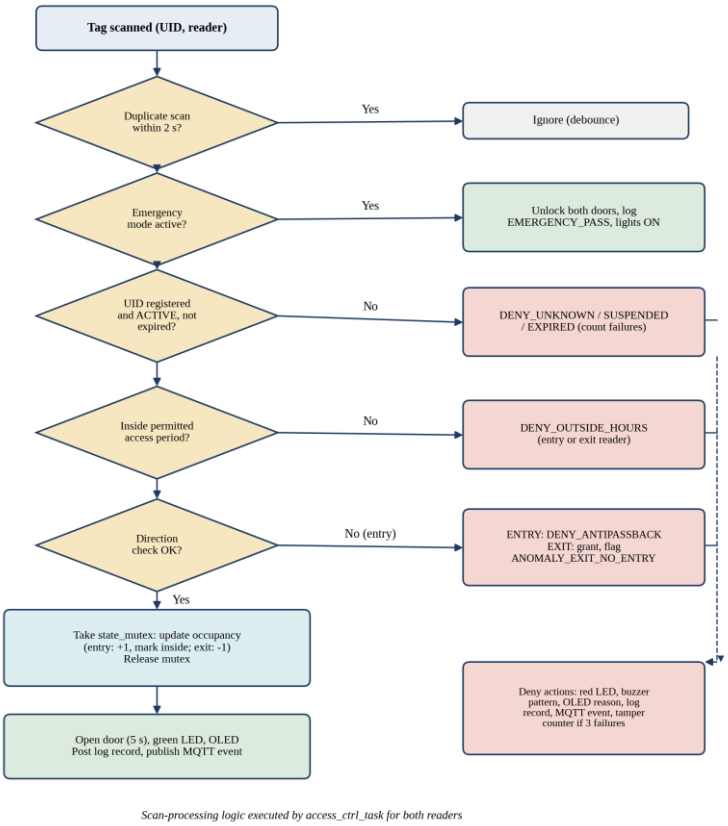


Figure 6: Scan processing and access decision flow

9.5 Occupancy and anti-passback model

- State per registered user: inside (boolean), entered_at (timestamp), last_scan (timestamp).
- ENTRY granted: inside = true, building count increments by one, entered_at set.
- EXIT granted: inside = false, building count decrements but never below zero.
- The occupancy table is protected by state_mutex and is snapshotted to NVS on every change and to SD every minute.
- On boot the Hub reloads the table from NVS, falls back to the SD snapshot, then reconciles with the last log lines.

9.6 Lighting state machine

Each room node implements the finite state machine in Figure 7. Transitions are driven by PIR events, the vacancy and warning timers, the LDR value with hysteresis, the override button and messages from the Hub (building occupancy and alerts).

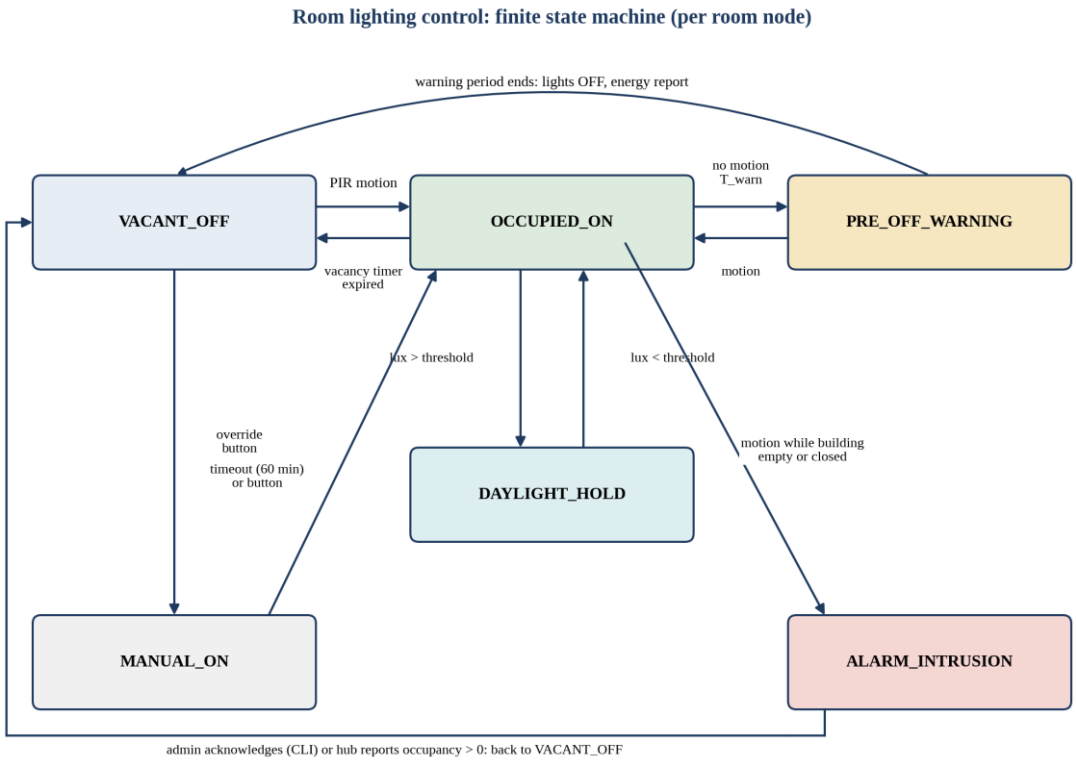


Figure 7: Lighting control state machine on a room node

Table 27: Lighting parameters and defaults

Parameter	Default	Demo value	Notes
Vacancy timeout (T_vac)	300 s	20 s	Time without motion before warning starts
Warning period (T_warn)	30 s	5 s	Lamp blinks at 1 Hz
PIR hold time	2 s	1 s	Ignore retriggers
Lux threshold on / off	600 / 800 (ADC counts)	Same	Hysteresis band prevents flicker
Manual override duration	60 min	30 s	Button press extends
Lamp power (for energy estimate)	72 W	72 W	Four 18 W tubes, configurable
Tariff	UGX 780 per kWh (configurable)	Same	Illustrative value, set by management

9.7 Operational timing parameters

Table 28: Timing and limit defaults

Parameter	Default
Door open time	5 s, then relock
Duplicate scan window	2 s
Heartbeat interval / offline threshold	5 s / 15 s
Distributed lock lease	3 s (auto-expire)
CLI session idle timeout	120 s
Login lockout	3 failed attempts then 60 s
Log flush	After each record and fsync every 10 s
Occupancy SD snapshot	Every 60 s
Task watchdog timeout	10 s

9.8 Reason and result codes

Table 29: Access result codes

Code	Meaning
GRANTED	Access allowed and door actuated
DENY_UNKNOWN	UID not in registry
DENY_SUSPENDED / DENY_EXPIRED	Registry status not ACTIVE or past expiry date
DENY_OUTSIDE_HOURS	Entry attempted outside the access period
DENY_EXIT_OUTSIDE_HOURS	Exit attempted outside the access period (while inside)
DENY_ANTIPASSBACK	Entry attempted while already recorded inside

Code	Meaning
ANOMALY_EXIT_NO_ENTRY	Exit granted with no recorded entry; flagged
EMERGENCY_PASS	Doors opened by emergency mode
ALERT_INTRUSION / ALERT_TAMPER / ALERT_BRUTE	Security alerts

9.9 Coding standards

- Language C (C99) with ESP-IDF conventions; one component per responsibility; public API in include/ headers.
- Every function that can fail returns esp_err_t; callers check with ESP_ERROR_CHECK only during init and handle errors explicitly at runtime.
- No blocking calls inside ISRs; use FromISR APIs and defer work to tasks. No malloc inside loops; allocate at init.
- All mutex takes use a timeout (pdMS_TO_TICKS) and the failure path is handled; follow the lock order cfg, state, sd.
- No magic numbers: configuration values live in Kconfig or a constants header. Log with ESP_LOGI, ESP_LOGW, ESP_LOGE and a per-module TAG.
- Each file begins with a comment block: purpose, author, requirement IDs covered.

10. Embedded Systems Concepts to Demonstrate

The assessment looks for a clear understanding of embedded systems principles. The table maps each concept to where it is used in this system, the ESP-IDF facility involved and the evidence a student can show in the video or presentation. Every team member should be able to explain at least the rows linked to their own work.

Table 30: Concept map

Concept	Application in this system	ESP-IDF facility	Evidence to show
GPIO, pull-ups and interrupts	Emergency, tamper, PIR, buttons	gpio_config, gpio_isr_handler_add	ISR posts to queue or event group
Debouncing	Buttons, duplicate RFID reads, PIR hold	esp_timer, tick comparison	Log shows one event per press
PWM	Servo position, buzzer tones	LEDC (ledc_timer, ledc_channel)	Servo angle in simulation
ADC and filtering	LDR daylight level	adc_oneshot, moving average	Lux values and hysteresis
SPI with multiple devices	SD card and two RFID readers on one bus	spi_bus_initialize, device CS	Bus guarded by sd_mutex
I2C	OLED and RTC on one bus	i2c_master	Two device addresses
UART	Serial CLI and diagnostics	uart driver, console	Interactive session
RTOS tasks, priorities, core affinity	Task set in Sections 9.2 and 9.3	xTaskCreatePinnedToCore	Task list with priorities
Queues	scan_queue, log_queue, door commands	xQueueSend, xQueueReceive	Producer-consumer diagram
Mutexes and priority inheritance	Occupancy, config, SD card	xSemaphoreCreateMutex	Lock order rule and timeouts
Binary semaphores and notifications	ISR to task signalling	xTaskNotifyFromISR	Latency measurement
Event groups	EMERGENCY, TAMPER, NET_UP, SD_OK flags	xEventGroupSetBits	Multi-condition waits
Software timers	Vacancy timer, warning timer, door relock	esp_timer, xTimerCreate	Timer callbacks
Finite state machines	Lighting FSM, door FSM, session FSM	Table-driven C code	State diagram matches code
Non-volatile storage	Credentials, schedule, occupancy	nvs_flash, nvs_set_blob	Value survives reboot
File systems on SD	Daily CSV logs, registry	esp_vfs_fat_sdspi_mount, FATFS	Files in SD image
Wi-Fi station mode	Network access	esp_wifi, esp_netif, event loop	Connect and reconnect logs

Concept	Application in this system	ESP-IDF facility	Evidence to show
MQTT (QoS, retained, Last Will)	Inter-node protocol	esp_mqtt_client	Topic trace
Time keeping	SNTP with DS3231 fallback, EAT conversion	esp_sntp, setenv TZ	Schedule uses local time
Watchdogs	Detect stuck tasks	esp_task_wdt	Forced hang is recovered
Cryptography	Password hashing, chained log	mbedtls SHA-256, PBKDF2	Verify command output
Memory management	Heap budget, stack sizing, ring buffers	heap_caps, high-water marks	Diagnostics command
Concurrency hazards	Race conditions, deadlock, priority inversion	Lock ordering, timeouts	Section 13 scenarios
Fault tolerance and fail-safe design	Degraded mode, store and forward	Event groups, retry back-off	Pull network, system continues
Hardware abstraction	RFID, door, sensors behind HAL	Component interfaces	Swap simulated reader
Configuration management	Kconfig, partitions	menuconfig, partition table	ROOM_ID per node

11. Inter-Node Communication Design

11.1 Transport

Nodes communicate with MQTT version 3.1.1 using the ESP-IDF esp_mqtt client. A public broker (for example broker.hivemq.com or test.mosquitto.org) is reachable from the Wokwi-GUEST network. Public brokers are not private, so topics use a unique group prefix and payloads avoid personal data (UIDs are sent as short hashes, names never leave the Hub). The code reads the broker URI and prefix from Kconfig so it can be changed in one place.

11.2 Topic structure

Let {g} be the unique group identifier chosen by the team (for example cocis26-g7). All topics are of the form {g}/...

Table 31: MQTT topics

Topic	Publisher	Subscriber	QoS / retain	Purpose
{g}/entrance/event	Hub	Rooms, tools	1 / no	Access event summary (granted, denied, anomaly)
{g}/entrance/occupancy	Hub	Rooms	1 / retained	Building count and open/closed state
{g}/room/{n}/telemetry	Room n	Hub	1 / no	Occupied flag, lamp state, lux, energy counters
{g}/room/{n}/event	Room n	Hub	1 / no	Motion, state changes, alerts
{g}/room/{n}/cmd	Hub	Room n	1 / no	Override, force off, test
{g}/config/schedule/prepare	Hub	Rooms	1 / no	Phase 1 of schedule update
{g}/config/schedule/ack	Rooms	Hub	1 / no	Acknowledgement of prepared version
{g}/config/schedule/commit	Hub	Rooms	1 / retained	Phase 2: activate version; late joiners read it
{g}/lock/request, grant, release	Any	Hub (coordinator)	1 / no	Distributed lock protocol (Section 13)
{g}/sys/emergency	Hub	All	1 / retained	Emergency on or off
{g}/node/{id}/status	Each node	Hub, tools	1 / retained	Heartbeat and Last Will (online or offline)

11.3 Message envelope

Every message is a compact JSON object with a common envelope so that handlers can validate and de-duplicate uniformly.

```
{
 "v": 1,                    // protocol version
 "id": "R1-0007-000142",    // node, boot count, sequence (unique)
 "ts": 1793088000,          // UTC epoch seconds
 "src": "R1",               // H, R1 or R2
 "type": "motion",          // message type
 "data": { "occupied": true, "lux": 412 }
}
```

Example payloads:

```
access event:   {"type":"access","data":{"uidh":"a1b2c3","dir":"IN","res":"GRANTED"}}
telemetry:      {"type":"tele","data":{"occ":true,"lamp":true,"lux":412,"wh":18.4}}
schedule prep:  {"type":"sched_prep","data":{"ver":12,"base":11,"days":[...],"hol":[...]}}
ack:            {"type":"sched_ack","data":{"ver":12,"ok":true}}
lock request:   {"type":"lock_req","data":{"res":"cfg","lease_ms":3000}}
```

11.4 Reliability of delivery

- QoS 1 (at least once) is used for every functional topic. Duplicates are expected, so every handler is idempotent and discards an already processed id.
- Retained messages on occupancy, schedule commit, emergency and node status give a late-joining or reconnecting node the latest state immediately.
- Each node sets a Last Will on its status topic so the broker announces an offline node without waiting for the heartbeat timeout.
- Outgoing messages that cannot be sent are placed in a bounded ring buffer (64 entries) and replayed in order after reconnect.

11.5 Time distribution

All nodes obtain UTC from SNTP and convert to East Africa Time. The Hub additionally holds a DS3231 RTC and uses it if SNTP is unavailable. Timestamps in messages are UTC epoch seconds, so ordering does not depend on local time zone settings. When a node has no valid time its events are marked with uptime-relative timestamps and corrected later by the Hub.

12. Data Design

12.1 Storage layout on the SD card

Table 32: SD card directory structure

Path	Content	Written by
/sdcard/config/users.csv	User registry	storage_task (on admin command)
/sdcard/config/schedule.json	Access schedule and holidays	storage_task (on commit)
/sdcard/config/settings.json	Thresholds, tariff, timeouts	storage_task
/sdcard/logs/access/YYYYMMDD.csv	Access events with hash chain	storage_task
/sdcard/logs/audit/YYYYMMDD.csv	Logins, configuration changes, alerts	storage_task
/sdcard/logs/energy/YYYYMMDD.csv	Per-room lamp-on time and energy estimates	storage_task
/sdcard/state/occupancy.json	Occupancy snapshot	storage_task (every 60 s)
/sdcard/export/	Reports generated by CLI export commands	storage_task

12.2 Record formats

Table 33: File schemas

File	Columns
users.csv	uid, name, role, status, expiry_date, created_at
access log	seq, ts_utc, ts_local, node, uid, name, direction, result, reason, prev_hash, hash
audit log	seq, ts_utc, actor, action, target, old_value, new_value, result, prev_hash, hash
energy log	ts_utc, room, lamp_on_seconds, baseline_seconds, est_wh_used, est_wh_saved
occupancy.json	version, ts_utc, count, inside[{uid, since}]

12.3 Schedule and settings model

```
{ "version": 12,
 "modified_by": "admin", "modified_at": "2026-10-30T09:14:00+03:00",
 "days": { "mon": ["07:00","21:00"], "tue": ["07:00","21:00"],
         "wed": ["07:00","21:00"], "thu": ["07:00","21:00"],
         "fri": ["07:00","21:00"], "sat": ["08:00","14:00"],
         "sun": null },
 "holidays": ["2026-12-25"] }
```

12.4 NVS namespaces

Table 34: NVS keys

Namespace	Keys	Purpose
auth	user_n_hash, user_n_salt, user_n_role, fail_n	Credentials and role (hash only)
sched	json_blob, version	Last committed schedule
occ	blob, version	Occupancy table
cfg	t_vac, t_warn, lux_on, lux_off, tariff, lamp_w	Tunable settings
sys	boot_count, seq_counter, last_hash	Event ID generation and log chain head

12.5 Hash-chain definition

For record i, hash_i = SHA-256(hash_(i-1) || canonical_text_of_record_i). The first record of each day starts from the last hash of the previous day, stored in NVS (sys/last_hash) and written in the file header. The verify command recomputes the chain and prints OK or the line number of the first mismatch.

12.6 Retention and failure handling

- Daily files keep individual files small and make rotation simple. Older files are retained until the card is 80 percent full, then the oldest energy logs are deleted first; audit and access logs are never auto-deleted.
- If a write fails, the record stays in the RAM ring buffer, the SD_OK bit is cleared and the system enters DEGRADED mode until the card is writable again.
- File writes are append-only; each record is flushed with fflush and fsync on a timer to balance safety and wear.

13. Synchronisation and Reliability Design

13.1 Hazards to design against

Table 35: Concurrency hazards and countermeasures

Hazard	Example in this system	Countermeasure
Race condition on occupancy	ENTRY and EXIT readers process two scans at the same instant and both update the count.	Single access_ctrl_task consumes scan_queue serially; state_mutex around the table.
Interleaved writes to the SD card	Access log, audit log and energy log written from different tasks corrupt a file.	Single-writer storage_task fed by log_queue; sd_mutex covers the shared SPI bus with the RFID readers.
Inconsistent configuration	Room 1 applies the new schedule while Room 2 still uses the old one.	Two-phase versioned update with ACKs (Figure 8); retained commit.
Duplicate events	QoS 1 redelivers an entry event; the count is incremented twice.	Unique event id and processed-id cache; idempotent handlers.
Lost update	Two admin sessions edit the schedule from the same base version.	Update carries base version; Hub rejects stale base.
Deadlock	Task A holds cfg waiting for sd; task B holds sd waiting for cfg.	Fixed lock order cfg, then state, then sd; timeouts on all takes.
Priority inversion	Low priority CLI task holds a mutex needed by access_ctrl_task.	Mutexes (not binary semaphores) with priority inheritance; short critical sections.
ISR and task sharing	ISR increments a counter while a task reads it.	Use FromISR APIs, atomic types or portENTER_CRITICAL; copy then process.

13.2 Local synchronisation primitives

Table 36: Primitives used on the Hub

Primitive	Protects or signals	Holder rule
cfg_mutex	Schedule and settings structure	Held only while copying or replacing the structure
state_mutex	Occupancy table and counters	Held during the single update step of a decision
sd_mutex	SPI bus and FAT operations	Held by storage_task for a write and by RFID tasks for a read transaction
scan_queue (depth 8)	Scan events from two readers	Producers never block longer than 20 ms; overflow is logged
log_queue (depth 32)	Records to be written	Drops and alerts if full for more than 1 s
system_event_group	EMERGENCY, TAMPER, NET_UP, SD_OK, TIME_OK	Set from ISR or tasks; read with xEventGroupWaitBits

13.3 Distributed synchronisation

Because three separate microcontrollers cannot share a mutex, consistency is obtained by three complementary techniques.

1. **Single-writer authority.** Only the Hub modifies the registry, occupancy, schedule and logs. Rooms propose changes through events or commands and never write shared records. This eliminates write-write conflicts.
2. **Versioned two-phase updates.** A configuration change is prepared on all nodes, acknowledged, then committed (Figure 8). Nodes keep serving the old version until the commit arrives, so no two nodes ever enforce different versions for longer than one network round trip. Late or rebooting nodes read the retained commit.
3. **Leased distributed lock.** For operations that need exclusive multi-node access (for example a global lights-off sweep at closing time or a log export), a node sends lock_req to the Hub coordinator. The Hub grants the lock to one requester at a time, each grant carrying a lease of 3 s and a fencing token that increases with every grant. A node that crashes while holding the lock loses it automatically when the lease expires, and stale holders are rejected by the token check.

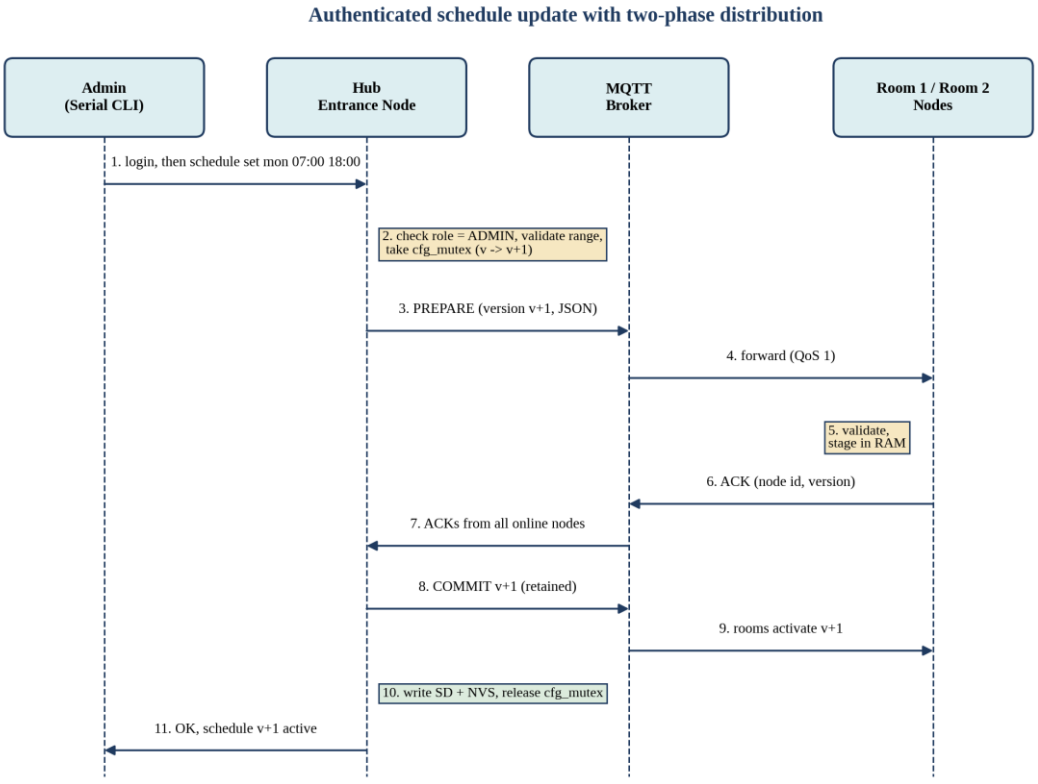


Figure 8: Authenticated schedule update with two-phase distribution

13.4 Failure modes and responses

Table 37: Failure analysis

Failure	Detection	System response
Wi-Fi or broker outage	MQTT disconnect event; heartbeat gaps	DEGRADED mode: local decisions, buffer events, retry with exponential back-off, blue LED slow blink.
Room node crash	Last Will or missing heartbeat for 15 s	Hub raises alert; room restarts under watchdog and resynchronises from retained messages.
Hub crash during scan	Task watchdog	Restart; occupancy reloaded from NVS; last log chain head restored; doors default to locked.
SD card removed or full	FAT error or free-space check	Ring buffer holds records, alert, resume on recovery.
Time not available	TIME_OK bit clear	Fail-secure: deny entry, allow emergency, show warning.
Sensor stuck (PIR always high)	No change for more than 2 hours	Diagnostics warning; lamp follows schedule limit.
Power loss	Brownout detector	State persisted on every change; doors lock on restart; mechanical release required for egress (documented).

14. Security Design

14.1 Authentication and roles

Table 38: Role permission matrix

Capability	VIEWER	SECURITY	ADMIN
View status, occupancy, rooms, lights	Yes	Yes	Yes
View access and audit logs	No	Yes	Yes
Roll-call, who is inside	No	Yes	Yes
Trigger or clear emergency mode	No	Yes	Yes
Suspend or reactivate a tag	No	Yes	Yes
Change access schedule and holidays	No	No	Yes
Add or remove users, change roles	No	No	Yes
Change thresholds, tariff, passwords	No	No	Yes
Verify and export logs	No	Yes	Yes

14.2 Credential and session handling

- Passwords are stored as PBKDF2-HMAC-SHA256 (at least 1000 iterations in simulation) with a 16-byte random salt per user. Plain text never touches NVS, SD or logs.
- The first boot creates a default ADMIN account with a one-time password that must be changed at first login.
- Three failed attempts lock the account for 60 s; each failure and lock is written to the audit log. Comparison uses a constant-time function.
- A session ends after 120 s idle or on logout. The prompt shows the current role, for example admin#.
- Optional INN-14: a physical admin action (card plus PIN on a keypad) is required for sensitive changes done at the door.

14.3 Threats and mitigations

Table 39: Threat analysis (STRIDE style)

Threat	Scenario	Mitigation
Spoofing	Cloned or shared tag	Anti-passback, suspension, alerts on repeated denials

Threat	Scenario	Mitigation
Tampering	Editing a log file to hide an entry	Hash-chained logs and verify command; audit of config changes
Repudiation	Admin denies changing schedule	Audit log with actor, time, old and new value
Information disclosure	Reading MQTT on a public broker	Group prefix, UID hashing, no names or secrets in payloads; TLS option documented for real deployment
Denial of service	Flooding scans or MQTT	Duplicate window, queue limits, rate limiting
Elevation of privilege	Viewer runs admin command	Role checks in the command dispatcher and a unit test per command
Physical tampering	Opening the reader cabinet	Tamper switch raises ALERT_TAMPER

> Security in the simulation is a proof of design. The report should state clearly which protections would be added in a real deployment (TLS to a private broker, secure boot, flash encryption, authenticated card protocols such as MIFARE DESFire).

15. Serial Interface and Management Menus

15.1 Interaction model

The CLI runs on UART0. A banner shows firmware version, time and mode. Commands are lower case words with arguments. Typing help lists the commands permitted for the current role; typing menu shows a numbered menu that maps to the same commands for users who prefer selection. Read-only commands work without login at VIEWER level; configuration commands require login.

15.2 Command reference

Table 40: Command set of the Hub CLI

Command	Role	Description
help, menu	All	List commands, show numbered menu
login <user>, logout, whoami	All	Authenticate (password prompt, no echo), end session, show role
status	VIEWER	Mode, time, schedule state, SD state, network state, uptime
occupancy	VIEWER	Building count and open or closed state
who	SECURITY	Names and UIDs of people inside, with entry time
rooms	VIEWER	Both rooms: occupied, lamp state, lux, vacancy timer
room <n>	VIEWER	Detailed status of one room including energy figures
lights	VIEWER	Lamp states and reasons (occupied, daylight hold, override)
log [n]	SECURITY	Last n access events (default 10)
log search <uid|name>	SECURITY	Filtered access events
log verify	SECURITY	Verify the hash chain; print first broken line
log export <date>	SECURITY	Create a report file in /sdcard/export
audit [n]	SECURITY	Last n audit entries
alerts	SECURITY	Active and recent alerts (intrusion, tamper, node offline)
nodes	VIEWER	Heartbeat status of the three nodes
schedule show	VIEWER	Current weekly schedule, holidays, version
schedule set <day> <open> <close>	ADMIN	Set a weekday window; example: schedule set mon 07:00 21:00
schedule off <day>	ADMIN	Close the building for a whole weekday

Command	Role	Description
schedule holiday add|del <date>	ADMIN	Maintain holiday list
user list	SECURITY	Registered users and status
user add <uid> <name> <role>	ADMIN	Register a tag
user suspend|activate|del <uid>	SECURITY (suspend), ADMIN (others)	Change status or remove
user temp <uid> <name> <hours>	ADMIN	Create a visitor card with expiry
config set <key> <value>	ADMIN	Change t_vac, t_warn, lux thresholds, tariff, door time
energy [room]	VIEWER	Lamp-on time, estimated kWh, UGX saved
emergency on|off	SECURITY	Start or end emergency mode; prints roll-call
release <uid>	ADMIN	Authorise a person to leave after hours (audited)
diag	VIEWER	Heap, stack high-water marks, queue depths, task list
sim scan <entry|exit> <uid>	ADMIN	Simulation only: inject an RFID scan
passwd	Own account	Change password

15.3 Example session

```
cocis> login admin
Password: ********
OK. Role ADMIN. Session expires after 120 s idle.
admin# schedule set mon 07:00 21:00
Preparing schedule v13 ... Room1 ACK, Room2 ACK ... committed.
admin# occupancy
Building: OPEN until 21:00 | Inside now: 3 | Room1: occupied, lights ON | Room2: vacant, lights OFF
admin# who
1) Okello J. (UID 04A1...) since 08:12   2) Nakato S. (UID 0B7C...) since 08:40   3) ...
```

15.4 Additional output interfaces

- OLED at the Hub: time, mode, building count, last event, alert banner.
- Status LEDs and buzzer patterns as in Section 8.5.
- MQTT status topics, which allow any MQTT client to subscribe and view live telemetry (useful during the demonstration).
- Optional: a minimal HTTP status page served by the ESP-IDF HTTP server (stretch goal).

16. Access Schedule Specification

16.1 Data and evaluation rules

1. The schedule maps each weekday to an open and close time, or to closed. A holiday date always means closed.
2. A time t is inside the period when open <= t < close in East Africa Time. Midnight-crossing windows are not supported; configure two days instead.
3. The Hub evaluates the rule for every scan using a copy of the schedule taken under cfg_mutex. Room nodes use their copy for intrusion and after-hours logic.
4. If the schedule version is unknown or time is invalid, the system treats the building as closed (fail-secure) and raises an alert.
5. A schedule change is committed only when all online nodes acknowledge. Offline nodes receive the retained commit on return.

16.2 Behaviour at and after closing time

- At close time the Hub switches to AFTER_HOURS, locks both doors against normal scans, and builds the list of people still inside (INN-10).
- The CLI and OLED display the list, security is alerted, and all lights in rooms with no motion are switched off.
- Exit scans outside the period are denied with DENY_EXIT_OUTSIDE_HOURS, as the brief requires (FR-SCH-03). An administrator can authorise a specific person to leave with the release command; the action is audited. Emergency mode always unlocks both doors for life safety.

> **Assumption A-1 (confirm with the lecturer):** the brief says individuals must not be permitted to leave the building outside the authorised period while still inside. The design implements this literally for normal scans, and adds two audited safety exceptions (administrator release and emergency mode). The report should present this as a deliberate safety-versus-policy decision.

16.3 Edge cases to test

- Scan exactly at the open and close second.
- Schedule edited while a person is mid-scan.
- Change from open to closed while people are inside.
- Clock not yet synchronised at boot.

- Holiday on a normally open weekday.

17. Simulation Plan on Wokwi

17.1 Day-one spike tests

The following small experiments decide the deployment option and part substitutions before any large coding begins. Each should take under an hour and must be recorded in the project log.

Table 41: Spike tests

#	Question	Pass criterion	If it fails
S1	Does an ESP-IDF project build and run in Wokwi?	Hello world and a FreeRTOS task print on the serial monitor	Use the Wokwi VS Code or CLI workflow with a local ESP-IDF build
S2	Can two Wokwi projects exchange MQTT messages through the Wokwi-GUEST network?	Messages published by A are received by B	Switch to deployment Option B
S3	Does SNTP return time in the simulation?	Correct UTC time printed	Use DS3231 RTC with manual set command
S4	Does the SD card part mount with FATFS and allow append and read?	File written and read back	Use NVS-only storage plus RAM ring buffer and document the limitation
S5	Which RFID source works?	A UID reaches the HAL from a part, custom chip or console	Console injection (sim scan command)
S6	Do servo, PIR, LDR, relay, OLED, RTC parts behave as expected?	Each part driven from the HAL	Replace with LED indicators and note in the report

17.2 Wokwi project set-up

- Create three projects: cocis-entrance, cocis-room1, cocis-room2 (room firmware is identical; ROOM_ID differs through sdkconfig).
- Keep diagram.json (circuit) and wokwi.toml in the repository so the circuit is reproducible; commit the final share links into docs/links.md.
- Use the pin maps in Section 8 exactly; any change must be updated in this document first.
- Prepare a demo sequence file (list of scan commands and PIR clicks) so the demonstration is repeatable under time pressure.

17.3 Demonstration scenarios for the presentation

Table 42: Scripted demo

#	Scenario	Shows
D1	Registered user scans ENTRY inside hours	Grant, door servo, OLED, log line, occupancy 1
D2	Same tag scans ENTRY again	Anti-passback denial
D3	Unknown tag scans ENTRY three times	Denials and brute-force alert
D4	Admin changes schedule so the building is closed, then a user scans	Authenticated configuration, propagation to rooms, outside-hours denial
D5	Person is inside when the building closes, tries to exit	Exit denial, roll-call list, administrator release
D6	Motion in Room 1	Lights on, status at the Hub, energy counter starts
D7	No motion	Pre-off blink, lights off, energy saved
D8	Bright LDR with motion	Daylight hold keeps lamp off
D9	Motion in Room 2 while building empty	Intrusion alert
D10	Press emergency button	Doors unlocked, lights on, roll-call
D11	Disconnect Wi-Fi on a room node	Offline operation, buffering, resynchronisation on reconnect
D12	Edit a log line then run log verify	Hash chain detects tampering

18. Testing and Validation

18.1 Strategy

Testing combines unit-level checks (pure logic such as the schedule evaluator, the hash chain and the lighting state machine, which can be run with scripted inputs), integration tests across nodes in the simulator, and a soak test. Every test case cites the requirement it verifies, and the results are pasted into the report as evidence.

18.2 Test cases

Table 43: Test case catalogue

ID	Test	Expected result	Verifies
TC-01	Valid tag at ENTRY inside hours	Door opens 5 s, green LED, log and count updated	FR-ACC-02, 03, 05; FR-OCC-01
TC-02	Unknown tag	Denied, red LED, DENY_UNKNOWN logged	FR-ACC-04
TC-03	Suspended and expired tags	Denied with correct reason	FR-ACC-08
TC-04	Duplicate scan within 2 s	Second read ignored	FR-ACC-06
TC-05	Anti-passback	Second ENTRY denied; EXIT without entry flagged	FR-ACC-07
TC-06	Three denied scans	Alert raised and logged	FR-ACC-09
TC-07	Who is inside	List matches entries minus exits	FR-OCC-02
TC-08	Reboot Hub with 3 inside	Same 3 users after restart	FR-OCC-03; NFR-04
TC-09	Motion in room	Lamp on within 1 s; Hub shows occupied	FR-LGT-01, 03; NFR-02
TC-10	No motion for T_vac	Pre-off blink then lamp off	FR-LGT-02, 04
TC-11	Bright LDR with motion	Lamp stays off (daylight hold)	FR-LGT-05
TC-12	Manual override	Lamp on for override time then normal	FR-LGT-06
TC-13	Hub offline, motion in room	Lamp still switches	FR-LGT-08
TC-14	Motion while building empty	Intrusion alert, lamp off	FR-LGT-09
TC-15	Schedule set without login	Rejected with message	FR-SCH-04, FR-CLI-04
TC-16	Schedule set as VIEWER or SECURITY	Rejected; audited	FR-SCH-04
TC-17	Schedule set as ADMIN	Both rooms ACK, version increments, entry rules change	FR-SCH-01, 06; FR-SYN-04
TC-18	Scan outside hours at ENTRY and EXIT	Both denied with correct reasons	FR-SCH-02, 03

ID	Test	Expected result	Verifies
TC-19	Emergency button	Doors unlock, lights on, roll-call printed	FR-SCH-08; INN-02
TC-20	Wrong password 3 times	Account locked for 60 s	NFR-05; INN-09
TC-21	Wi-Fi drop on a room	Offline flag in 15 s; events replayed on return without duplicates	FR-DST-03, 04; FR-SYN-03
TC-22	Remove SD card	DEGRADED mode, buffer, recovery on reinsert	FR-STO-04
TC-23	Edit log line, run log verify	First broken line reported	FR-STO-05; INN-03
TC-24	Concurrent scans at both readers plus CLI commands	No corruption, no negative count, no deadlock	FR-SYN-01, 07
TC-25	Stale schedule update (old base version)	Rejected	FR-SYN-04
TC-26	30 minute soak with random scans and motion	No crash; heap and stack margins hold	NFR-03, 08

18.3 Acceptance criteria

- All Must requirements pass their test cases in the simulator.
- At least the Tier 1 innovations (INN-01 to INN-06, INN-08, INN-09) are demonstrated live.
- Each member can run their own feature in the demo and explain the code and the concepts used.
- The report, simulation links, videos, contribution statements and member details are submitted before the deadline.

19. Project Management

19.1 Suggested work breakdown for five members

The roles below spread the work so that every member owns a distinct, explainable part. Each member must also review at least one other member's code, since marks are lost when a student cannot explain a claimed contribution. If the team is smaller, merge roles 4 and 5 first.

Table 44: Roles and responsibilities

Role	Main ownership	Key deliverables	Concepts to master for Q&A
M1: Access and RFID lead	Hub hardware, RFID HAL, doors, access decision, occupancy, anti-passback, emergency	FR-ACC, FR-OCC, INN-01, INN-02, INN-12	SPI, GPIO ISR, PWM, queues, FSM
M2: Storage and integrity lead	SD driver, log writer, CSV, hash chain, NVS persistence, energy logs	FR-STO, INN-03, INN-07	FATFS, VFS, mbedtls, NVS, single-writer design
M3: Room and lighting lead	Room hardware, PIR, LDR, relay, lighting FSM, energy counters, intrusion detection on room side	FR-LGT, INN-04 (room), INN-05, INN-06	ADC, timers, hysteresis, state machines
M4: Networking and synchronisation lead	Wi-Fi, MQTT, SNTP, protocol, distributed lock, versioned updates, offline buffer, heartbeat	FR-DST, FR-SYN, INN-08, INN-13	MQTT QoS, Last Will, idempotency, locks, race conditions
M5: CLI, security, schedule and documentation lead	CLI, authentication, RBAC, schedule engine, tests, report assembly	FR-CLI, FR-SCH, INN-09, INN-10, test log, report	UART, parsing, crypto, RBAC, time zones

19.2 Timeline (27 October to 2 November 2026)

Table 45: Day-by-day plan

Date	Goals	Exit criterion
Tue 27 Oct	Kick-off; read this document; create repository and group MQTT prefix; run spike tests S1 to S6; assign roles	Deployment option and RFID source decided
Wed 28 Oct	HAL drivers and Wokwi circuits for all nodes; shared types; MQTT hello between nodes; storage skeleton	Every peripheral works from a test task
Thu 29 Oct	Access engine, occupancy, lighting FSM, schedule engine, event protocol, SD logging; first end-to-end demo D1 to D7	Core Must requirements work together
Fri 30 Oct	CLI, authentication, schedule distribution, Tier 1 innovations, hash chain, emergency	TC-01 to TC-20 pass
Sat 31 Oct	Integration, fault tests, soak test, bug fixes, feature freeze at evening; report drafting by owners	TC-21 to TC-26 pass; code frozen

Date	Goals	Exit criterion
Sun 1 Nov	Record individual videos; finalise report with figures and test evidence; publish Wokwi links; rehearse Q&A	Report and videos ready
Mon 2 Nov	Final check, submit on Muele before the deadline, present to lecturer	Submission confirmed

19.3 Working agreements

- Version control: one Git repository, short-lived feature branches, pull requests reviewed by a second member, commit messages that cite requirement IDs.
- Definition of done for any feature: code merged, requirement ID in header comment, test case passed and recorded, demo step written, owner able to explain it without notes.
- Interface changes (topics, payloads, pins, structs) require an update to this document in the same pull request and notification to the team.
- Daily 15 minute stand-up: yesterday, today, blockers. Keep a shared risk and decision log.

20. Report, Video and Presentation Guide

20.1 Recommended report structure

Table 46: Report outline mapped to the brief

Chapter	Content	Reuse from this document
1. Introduction	Problem, objectives, scope	Sections 1 to 3
2. Requirements	Functional, non-functional, innovative features	Sections 4 to 6
3. System architecture	Overall design, node roles, communication, deployment on Wokwi	Section 7, 11
4. Hardware design	Components, pin maps, wiring figures	Section 8
5. Software design	Tasks, state machines, algorithms, data and storage	Sections 9, 12, 15, 16
6. Synchronisation and security	Mutexes, distributed protocol, authentication	Sections 13, 14
7. Implementation and operation	How to run, screenshots of CLI and logs, demo walkthrough	Section 17
8. Testing and results	Test table with evidence	Section 18
9. Innovation summary	What was added and why it matters	Section 6
10. Contributions and member details	Individual contributions, names and identification details	Section 19 and group table
11. Conclusion and future work	Limits, lessons, real deployment recommendations	Section 21
Appendices	Wokwi links, video links, code excerpts	Section 20.3

20.2 Individual video checklist

- Length 5 to 8 minutes, screen recording with voice, face optional.
- Start by stating name, student number, role and the requirement IDs you implemented.
- Show the code you wrote file by file, explaining data structures, tasks, synchronisation and error handling.
- Run the simulation and show your feature working, including one failure case.
- Name the embedded systems concepts used (use Section 10) and why you chose them over alternatives.
- Do not read from a script without understanding; you will be asked questions on the same material.

20.3 Submission checklist

Table 47: Final checklist

Item	Done
Report with all required sections, figures and test evidence	
Wokwi link(s) open without login and run	
One video link per group member, accessible to the lecturer	
Individual contribution statement for every member	
Names and identification details of all members	
Submitted on Muele before 2 November 2026	
Every member available for the presentation	

20.4 Likely questions in the presentation

- Why a hub-and-spoke design, and what happens when the Hub fails?
- Show me where a race condition could occur and which primitive prevents it.
- What is priority inversion and how does your mutex configuration address it?
- Why is only one task allowed to write to the SD card?
- How does your system guarantee that all rooms use the same schedule version?
- How are passwords stored, and what would you change for a real deployment?
- Why use QoS 1 and what problem does that create?
- How did you choose task priorities and stack sizes?
- What happens to occupancy when power is lost during a scan?
- Explain your lighting state machine and how you avoid flicker.

21. Risks and Mitigations

Table 48: Risk register

ID	Risk	Likelihood	Impact	Mitigation
R1	Wokwi lacks an RFID part	High	Medium	HAL with simulated reader (console or custom chip); decide on day one
R2	Multi-project MQTT does not work in the simulator	Medium	High	Spike S2; fall back to Option B with the same message design
R3	Six days is too short for full scope	High	High	Strict Must-first order; Tier 2 and 3 only after Tier 1 is stable; daily integration
R4	SD card simulation limitations	Medium	Medium	Spike S4; NVS and ring buffer fallback documented
R5	Public broker unreliable or shared with other groups	Medium	Medium	Unique topic prefix; retry logic; local fallback
R6	Integration bugs from late merging	High	High	Shared types component, daily merges, protocol tests with recorded messages
R7	A member cannot explain their code	Medium	High	Peer reviews, own-feature demos, rehearsal of Q&A
R8	Race conditions appear only late	Medium	Medium	Design rules in Section 13; stress test TC-24; code review checklist
R9	Simulator timing differs from real hardware	Low	Low	State assumption in the report; use timeouts and retries, not fixed delays

22. Guidelines for AI Coding Agents

This section is addressed to AI agents that read this document to generate, review or document code for the team. Following these rules keeps contributions consistent and safe to merge.

1. **Read order:** Sections 1.4 (conventions), 4 (requirements), 8 (pins), 11 (protocol), 12 (data), then the section relevant to the assigned task.
2. **Cite requirement IDs** in header comments, commit messages and test names. Do not invent new IDs; propose additions to the team.
3. **Never change** GPIO assignments, topic names, JSON field names, reason codes, schedule or NVS keys without an accompanying update to this document and approval from the task owner.
4. **Use ESP-IDF (C) and FreeRTOS APIs only.** Do not generate Arduino code. Target the ESP32 and ESP-IDF 5.x style APIs; check the version used by the Wokwi project before using newer driver APIs.
5. **Respect the architecture:** hardware access only through HAL components; only storage_task writes the SD card; only the Hub modifies authoritative data; lock order cfg, state, sd; every mutex take has a timeout.
6. **Handle errors explicitly,** log with ESP_LOG macros, and keep ISRs minimal.
7. **Do not hard-code secrets.** Default credentials come from Kconfig and must be changed at first login.
8. **Produce testable units:** keep pure logic (schedule evaluator, hash chain, state machine step function, JSON codec) free of hardware calls so it can be tested on the host.
9. **State assumptions and open items** in comments marked TODO(ID) rather than guessing silently, especially for simulator part behaviour that is still unverified.
10. **Style of prose output:** use plain ASCII hyphens, never long dashes, and keep explanations short and factual.

Appendix A: Glossary

Table 49: Terms and abbreviations

Term	Meaning
ADC	Analog to digital converter
Anti-passback	Rule that prevents a tag from entering twice without an exit between
CLI	Command line interface
EAT	East Africa Time, UTC+3
ESP-IDF	Espressif IoT Development Framework
FATFS	File allocation table file system library
FSM	Finite state machine
HAL	Hardware abstraction layer
Idempotent	Operation that has the same effect if applied more than once
LDR	Light dependent resistor
LEDC	ESP32 PWM peripheral
LWT	MQTT Last Will and Testament
MQTT	Message Queuing Telemetry Transport, publish and subscribe protocol
NVS	Non-volatile storage in flash
PBKDF2	Password-based key derivation function
PIR	Passive infrared motion sensor
RBAC	Role-based access control
RFID	Radio frequency identification
SNTP	Simple network time protocol
TWDT	Task watchdog timer
UID	Unique identifier of an RFID tag

Appendix B: Requirement Traceability Matrix

Table 50: Requirement areas to design sections and tests

Brief section	Requirement IDs	Design sections	Test cases
2.1 Access control and occupancy	FR-ACC-01 to 09, FR-OCC-01 to 06	7, 8.2, 9.4, 9.5	TC-01 to 08, 24
2.2 Automated lighting	FR-LGT-01 to 09	8.3, 9.3, 9.6	TC-09 to 14
2.3 Distributed ESP32 control	FR-DST-01 to 05	7, 11	TC-21
2.4 Data storage	FR-STO-01 to 06	12	TC-08, 22, 23
2.5 Serial interface	FR-CLI-01 to 06	15	TC-07, 15, 20
2.6 Configurable schedule	FR-SCH-01 to 08	16, 14	TC-15 to 19, 25
2.7 Synchronisation and reliability	FR-SYN-01 to 07	13	TC-21, 24, 25, 26
3 Innovation emphasis	INN-01 to 15	6	TC-05, 14, 19, 20, 23

Appendix C: Default Configuration Values

```
door_open_ms       = 5000
dup_scan_ms        = 2000
t_vac_s            = 300      (demo: 20)
t_warn_s           = 30       (demo: 5)
override_min       = 60       (demo: 0.5)
lux_on / lux_off   = 600 / 800 (ADC counts, 12-bit, tune in simulation)
heartbeat_s        = 5        offline_after_s = 15
lock_lease_ms      = 3000
session_idle_s     = 120      login_fail_max = 3   lockout_s = 60
lamp_w             = 72       tariff_ugx_per_kwh = 780
tz                 = EAT-3
mqtt_prefix        = <unique group id>
```

Appendix D: Open Questions for the Lecturer

1. Is it acceptable to demonstrate three terminals as three linked Wokwi projects, or is a single project expected?
2. Is the literal interpretation of the exit restriction (Assumption A-1) acceptable, with the audited release and emergency exceptions?
3. Is a simulated RFID source (console or custom chip) acceptable if no RFID part is available?
4. Is an external MQTT broker acceptable for inter-terminal communication in the simulation?
5. May the room nodes share the Hub's SD card through the Hub, or must each terminal have its own memory card module?